



HWA CHONG INSTITUTION
C1 Promotional Examination
Higher 2

NAME

CT GROUP

20S

CHEMISTRY

9729/03

Paper 3 Free Response

1 October 2020

55 min

Candidates answer on the Question Paper.

Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper. If additional space is required, you should use the pages at the end of this booklet. The question number and part must be clearly shown.

Section A

Question 1 is a compulsory question.

Section B

Answer only **one** question from this section.

The use of an approved scientific calculator is expected, where appropriate.

A Data Booklet is provided.

You are reminded of the need for good English and clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

No. of sheets of writing paper submitted (write 0 if none)	
Calculator Model	

For Examiner's Use	
Question No.	Marks obtained
1	/ 20
Circle your option below	
2 / 3	/ 15
Deductions	
Total Marks	35

Section A

Question 1 is a **compulsory** question. All candidates should answer this question.

- 1 The peroxides are a group of compounds containing an O–O single bond. Many peroxides are known to be oxidising agents.

- (a) Barium peroxide, BaO_2 , is an inorganic peroxide. Draw the dot-and-cross diagram of the ionic compound BaO_2 . [1]

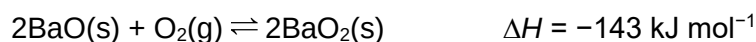
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- (b) Barium peroxide, BaO_2 , can be synthesised by mixing barium oxide, BaO , with oxygen gas at about 500 °C.



- (i) State *Le Chatelier's Principle*. [1]

- (ii) Write the expression for the equilibrium constant, K_p , for this reaction. [1]

- (iii) State and explain the effect on the position of this equilibrium when

- (I) the temperature is increased to 600 °C;
 (II) some $\text{BaO}(\text{s})$ is added. [3]

- (iv) In each of the cases in (b)(iii), state how the value of K_p will change. [2]

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- (c) A mixture of BaO(s) and $\text{O}_2\text{(g)}$ was placed in a vessel and allowed to reach equilibrium at $500\text{ }^\circ\text{C}$. The partial pressure of $\text{O}_2\text{(g)}$ was found to be 1.05 atm at equilibrium.

The volume of the vessel was then halved and the reaction allowed to equilibrate at the same temperature.

Deduce the partial pressure of $\text{O}_2\text{(g)}$, in atm, at the new equilibrium. Show your reasoning. [2]

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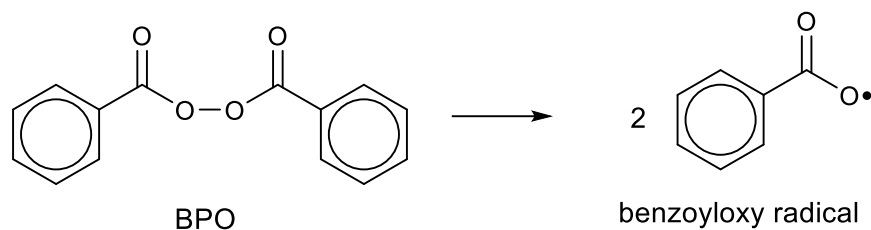
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- (d)** Benzoyl peroxide (BPO) is an organic peroxide found in acne medication like OXY5™. Its medical efficacy is based on the decomposition of BPO to form benzoyloxy radicals.



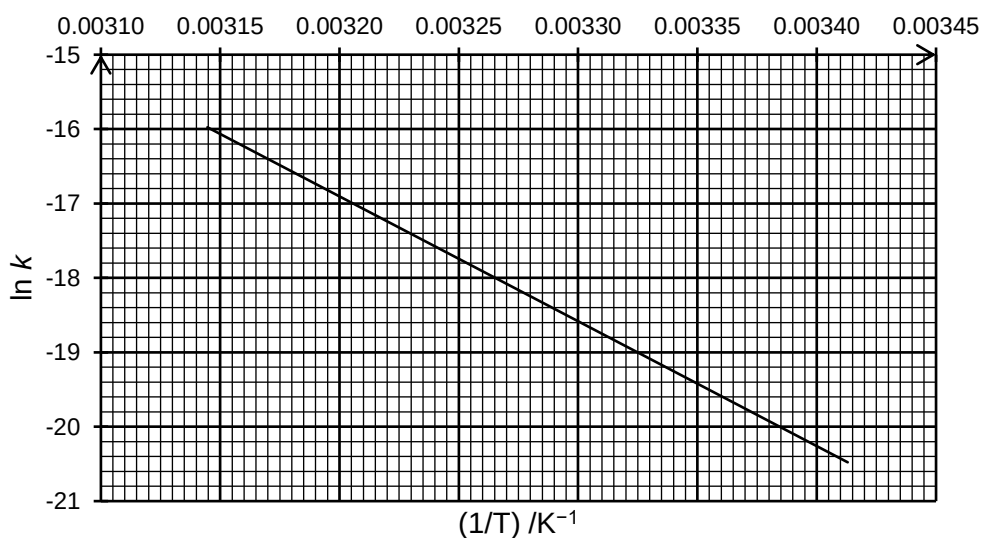
- (i) Explain what is meant by the term *free radical*. [1]
- (ii) Copy the structure of BPO and draw curly arrows to illustrate how the bond in BPO breaks to give benzoyloxy radicals. [1]

The equation below shows how the rate constant, k , for the decomposition of BPO is affected by temperature, T in Kelvin, and activation energy, E_a .

$$\ln k = -\frac{E_a}{RT} + \ln A$$

A is the pre-exponential factor, a constant, and R is the molar gas constant.

To investigate the decomposition of BPO, the rate constant k (units in s^{-1}) was determined at various temperatures in an experiment. A graph of $\ln k$ vs $1/T$ was plotted and shown below.



- (iii)** Using the graph, determine the activation energy, E_a , for the decomposition of BPO. [2]

A sample of OXY5™ contains 5% by mass of BPO and is stored at a temperature of 37 °C for 60 days. The decomposition of BPO is known to follow first-order kinetics.

- (iv) Using the graph, determine the rate constant, k , for the decomposition of BPO at 37 °C. [2]
- (v) Determine the half-life, in seconds, of the reaction at 37 °C. [1]

- (vi) The concentration of a reactant, $[R]$, after n half-lives can be determined using the equation below

$$[R] = [R]_0 \times \left(\frac{1}{2}\right)^n$$

where $[R]_0$ is the initial concentration of the reactant.

Use your answer to (d)(v) to determine the percentage by mass of BPO left after 60 days. [2]

- (vii) Suggest a method whereby the shelf-life of OXY5™ could be increased. [1]

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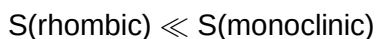
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2 (a) (i) Define the term *standard enthalpy change of formation of calcium sulfate*. [1]

enthalpy change of atomisation of Ca(s)	= +155 kJ mol ⁻¹
lattice energy of calcium sulfate	= -2489 kJ mol ⁻¹
S(s) + 2O ₂ (g) + 2e ⁻ → SO ₄ ²⁻ (g)	= -839 kJ mol ⁻¹

[illegible]

(b) At 366 K, two allotropes of sulfur, rhombic and monoclinic, exist in dynamic equilibrium.



- (i)** Given that the ΔS of the conversion from S(rhombic) to S(monoclinic) is $+0.82 \text{ J mol}^{-1} \text{ K}^{-1}$, calculate the ΔH of this conversion at 366 K. [1]
- (ii)** Using your answer to **(b)(i)**, explain which allotrope of sulfur, rhombic or monoclinic, would have a more negative enthalpy change of combustion. [2]

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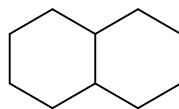
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- (c) Alkane **W** is heated with bromine at 300 °C to form three different monobrominated products which are constitutional isomers of one another.

**W**

- (i) Draw the **skeletal formulae** of the three monobrominated products, ignoring stereoisomerism. [3]
- (ii) Predict the ratio in which these three products are formed in the reaction. You may assume that all the hydrogen atoms in **W** are replaced by bromine atoms at the same rate. [1]

Only **two** of the monobrominated products exhibit enantiomerism because of the presence of chiral centre(s).

- (iii) Explain what is meant by a *chiral centre*. [1]
- (iv) Indicate, with asterisks (*), all chiral centres present in the compounds you have drawn in (c)(i). [2]

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[Total: 15]

- 3 Methane has important value as a fuel and as a chemical feedstock, in place of petroleum, due to its availability in large amounts from common sources.

(a) Why is it important to seek alternatives to petroleum? [1]

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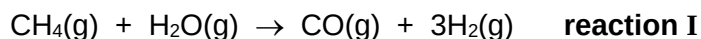
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(b) For decades, the industry indirectly converts methane to hydrocarbons using the following method.

Methane is first reacted with steam to form 'synthesis gas', a mixture of carbon monoxide and hydrogen.



CO is then converted to longer chain hydrocarbons.

(i) What is meant by the term *bond energy*? [1]

(ii) Using relevant bond energies from the *Data Booklet*, calculate the enthalpy change for reaction I. [2]

(iii) Deduce, with reasoning, the sign of ΔS for reaction I, and explain why a high temperature is required for the reaction to be spontaneous. [2]

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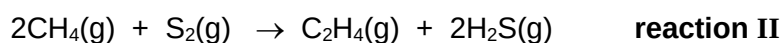
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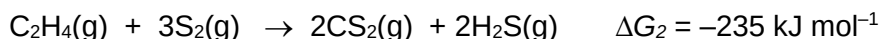
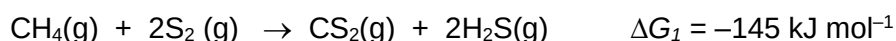
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- (c) The method above is expensive, hence research on direct conversion of methane to ethene has drawn tremendous efforts since 1980s.

A recent article in Nature journal describes the use of sulfur to make alkenes directly from methane. At 1323 K, sulfur exists as $S_2(g)$, which reacts with methane over a solid catalyst:



- (i) What is the role of sulfur in reaction II? [1]
- (ii) Using the following information, draw an energy cycle to calculate the Gibbs' free energy change of reaction II at 1323 K.



ΔG_1 and ΔG_2 are values at 1323 K. [3]

- (iii) The mechanism of reaction II involves the formation of C–S and H–S bonds between methane and sulfur with the solid catalyst.

Describe **two** ways in which the heterogeneous catalyst provides an alternative pathway of lower activation energy to increase the rate of reaction II. [2]

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(d) Methane and other alkanes can also form useful compounds via free radical substitution, which introduces halogen atoms that can further convert to other functional groups.

(i) Write the **two** propagation steps in the free radical substitution of methane with bromine to form bromomethane. [2]

(ii) Trace amounts of ethane is detected in the above reaction. Draw the corresponding hydrocarbon product expected if 2,2-dimethylpropane were used instead of methane. [1]

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[Total: 15]

~END OF PAPER~

Additional answer space

If you use the following pages to complete the answer to any question, the question number must be clearly shown.

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